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A 'String Figure' Perspective on Heritage Sites - Examining a Village, a Fishing Harbour and a Former Railway Area in Denmark

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Abstract

This paper will take its departure from research conducted by the author at Aarhus School of Architecture concerning the Danish heritage category 'cultural environments'. This heritage category concerns heritage sites and incorporates both buildings, structures and landscapes. In other words; 'cultural environments' are built wholes. The research has focused on how to manage these sites in a sustainable manner and this has led to new insights into the connection between the intangible and tangible aspects. The paper will be based on this research, but further the discussion and reflection through the so-called 'string figures' described by professor Donna Haraway (2016). The paper will analyze three Danish heritage sites; a small village in the countryside, an historic fishing harbor in a small town and a former railway area in the city centre and pair these places with the 'string figure' perspective as a method to deepen our understanding of the places. The connections between the different physical elements, natural elements, people and more-than-humans can provide a new understanding of what is important to care for within the historical site. Furthermore, the theory behind the 'string figures' outlines an approach to a more detailed narrative of the site which elaborates on the historical relevance and the deep meanings that the site contains. This way of understanding the many layered historical sites creates a direction for how to manage and care for the sites in the future. The historical buildings do not stand alone, and their cultural, historical relevance is often interconnected with the natural surroundings, along with plants and animals, and the people that have cared for the buildings through history and today. Donna Haraway's 'string figures' show the importance of the connections, and not the isolated objects or buildings, which can create a new understanding of our heritage sites.

Keywords: *heritage; Haraway; ecology; connections; methods; management*

1. Introduction

The paper will analyze three different historical sites by mapping the tangible and intangible components as well as the dynamic and static aspects of the sites. However, the analysis will not focus on these aspects and components as isolated entities, but rather the connections they form. This is investigated through professor Donna Haraway's, so-called, "string figures" (Haraway, 2016). The three sites are selected on the basis of an extensive research work conducted at Aarhus School of Architecture from 2015-2022 where over 2300 *cultural environments* in Denmark have been mapped and described (Morgen et al., 2025). The term

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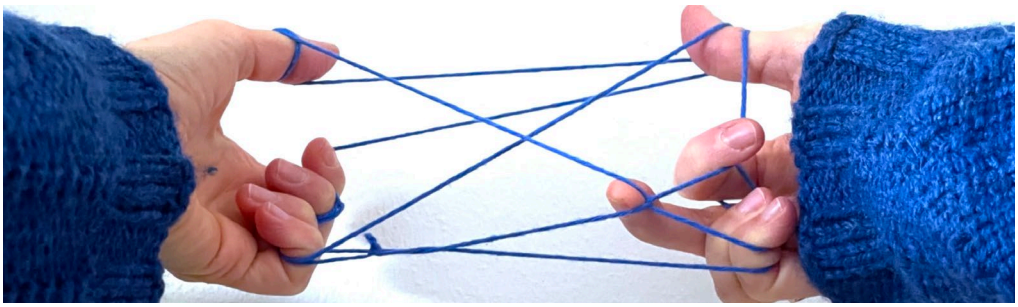


Fig. 1 – *Picture of the string figure game or "cat's cradle" performed by the author* (Kirkegaard, 2025).

cultural environments are a Danish planning category where a collection of buildings, built structures and the landscape, in unison, conveys a cultural, historical narrative. Within the official Danish management of cultural environments, the following description is used to evaluate whether a site is a cultural environment: The site should be *"*a built whole where significant aspects of the societal development can be experienced*"* (Skov- og Naturstyrelsen, 1999, p. 7). Cultural environments are a heritage category that both can comprise urban areas and in rural areas. The three sites, this paper will describe, are: A small village in the countryside, called Alling, an old fishing harbor, called Ebeltoft Fishing Harbor and a former railway area, called Godsbanen, in the city centre of Aarhus. Ebeltoft Fishing Harbor, Alling and Godsbanen has been the subject of investigation in the research project *Sustainable Cultural Environments* (2022-2026) (Kirkegaard et al., 2025) and the Ph.D. thesis *Transformation of Cultural Environments* (2017-2021) (Kirkegaard 2021) at Aarhus School of Architecture.

2. String Figures

The American professor emerita Donna Haraway describes a practice in her book *Staying With the Trouble – Making Kin in the Chthulucene* from 2016 by two letters: "SF". The "SF" stands for "science fiction", "speculative fabulation", "string figures", "speculative feminism", "science fact" and "so far". This "SF"-practice is an attempt to approach our world through modesty, mixed methods and an acknowledgement of the world's complexities (Haraway, 2016, p. 10). Donna Haraway works within and in-between different disciplines, such as art, philosophy, science and biology. One could argue that her work is a part of the new ecological discourse, along with Bruno Latour and Anna Tsing, among other scholars. Some of Haraway's references in her theory are to the English mathematician and philosopher Alfred North Whitehead whose theory, e.g. *Adventures of ideas* from 1933, has been a base for ecological theory. Whitehead argues that relevant things in nature are the outcome of their interconnections and this interconnectedness shows us how everything has internal relations to other things (Whitehead, 1933, p. 201). A string figure can also be called "cat's cradle" and is a string game you play with your hands (Fig. 1). It is something that kids can play, but it has also been a means of telling stories within native communities. The string runs from finger to finger in a pattern that forms a "picture" or a story. In the string game the string is passed to another participant in the game. When given to another participant the string figure changes and so the story will change.

Haraway uses the *string figure* as an analogy for the connections between what she calls "critters": microbes, plants, animals, machines, humans, and nonhumans. Within the *string figure* the "critters" are an analogy for the fingers, so the string is the connector between the fingers, or the, so-called, "critters".

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To exemplify this view: The farmer is a critter, his cow is a critter and his field is a critter and these three critters are interconnected in the *string figure* by strings forming a pattern or a "figure". Haraway describes how: "String figures can be played by many, on all sort of limbs" and she continues; "as long as the rhythm of accepting and giving is sustained" (Haraway, 2016, p. 10). Haraway explains that sometimes you drop a sting and other times you form something that was not there before. In the following description, I will describe three cultural environments through this *string figure* perspective by mapping the sites' different "critters": people, nature, and buildings, and span connections between them as a means to nuance the narrative of the cultural environment and form directions for future developments.

3. Three *string figure* heritage cases

3.1. Introducing Ebeltoft Fishing Harbor through the *string figure* perspective

I will describe Ebeltoft Fishing Harbor (Figs. 2 & 3) through the lens of the *string figure* perspective. I have concluded that the cultural environment has seven, so-called, critters: 1) *The Buildings*, 2) *The Urban Layout*, 3) *The Leisure Associations*, 4) *The Businesses*, 5) *The fish*, 6) *The sea* and 7) *The visitors*. The critters are a mixture of physical objects, animals, nature, people and activities (that are tied to associations or businesses and thus consists of both objects and people). Another researcher could point to other critters or an even larger array of critters, but as Haraway states: "It matters what thoughts thing thoughts" (Haraway, 2016, p. 35) and in this case I am forming the *string figure* as an urban planner and researcher in cultural environments. Thus, the narrative, that the *string figure* will form, will be different if another person is creating it. However, the relevance of the *string figure* lies within its ability to shed light on connections within the environment which may be hidden or underrepresented when managing the site in a planning and heritage context.



Fig. 2 – *Picture of one of the sheds at Ebeltoft Fishing Harbor, Denmark* (Kirkegaard, 2019).



Fig. 3 – *Ebeltoft Fishing Harbor has a 125-year-old fishing association and one industrial fishing boat left. The picture shows the fishing boat at Ebeltoft Fishing Harbor, Denmark* (Kirkegaard, 2019).

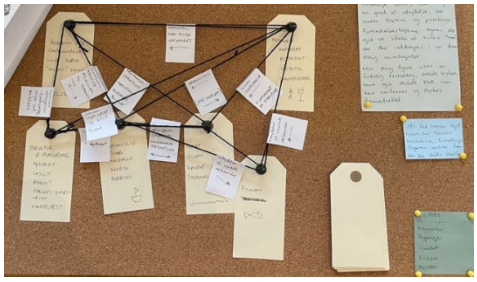


Fig. 4 – *Example of a string figure analysis with strings, connections, and critters, needles with notes stating the critter* (Kirkegaard, 2025).

The connections at Ebeltoft Fishing Harbor are many (around 20 connections) and some of these connections seem symbiotic and others expresses an exploitation (Fig. 4). The fishing harbor has historically been a place of trade and fishing. The fishing industry has been, and is still, in decline and currently the harbor holds one fishing company with one boat. Besides the one fishing boat, the harbor has been a place for leisure

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activities concerning boats and fishing, but recently new businesses have moved into vacant buildings. These businesses are: A wine shop, an Italian delicatesse shop, a glass art shop, a champagne bar, a café, a restaurant and a furniture store (Kirkegaard 2021, pp. 232). I have noted these businesses as one overarching critter called "The Businesses" in my diagram and this critter forms a connection to the following other critters: 1) The Buildings, 2) The Urban Layout, 5) The Fish, 6) The Sea and 7) The Visitors.

The Businesses are connected to *The Urban Layout* by their presence in the shared, public space with furniture, people and activities in the course of a day, week and year. *The Businesses* and *The Urban Layout* form a connection that is important for the identity of the heritage site and has consequences for the space left for the maritime leisure associations and it strongly impacts what type of visitors that are visiting the site. After I conducted interviews and observed the site during approximately two years (Kirkegaard, 2021) it became clear that within the public space there was a conflict between *The Businesses* and *The Leisure Associations*, which felt pressured by the unspoken guidelines formed during the years of inhabiting the space and due to the physical spaciousness of the shared space. The *string figure* does not form a connection between *The Leisure Associations* and *The Businesses* since there are none. However, for future sustainable management a connection between those two could result in less social friction at the site. In other words, a more symbiotic connection between the two could help sustain *The Leisure Associations* which is pressured by the forces of *The Businesses*. It is important to note that *The Leisure Associations* is "doing" the intangible heritage, such as fishing and boat repair. This intangible, but site-specific, heritage carried out by *The Leisure Associations* is often overlooked in planning and heritage management and easily destroyed by a lack of understanding of which other aspects and forces that impacts it.

The focus on the *connections,* both those that show a symbiotic relationship and connections that are lacking or resulting in a less harmonious environment between people, structures, the layout, buildings, nature and/or animals, are one of the abilities of the *string figure* perspective. The *string figure* analysis for Ebeltoft Fishing Harbor also sheds light on three other connections that is worth paying attention to: Firstly, the connections between *The Fish* and *The Businesses*, secondly the connection between *The Fish* and *The Sea,* and thirdly the connection between *The Fish* and *The Leisure Associations*. *The Fish* covers the underwater animals as a whole; molluscs, clams, porpoise, cod, and all other sea creatures that are a part of the Danish marine life ecosystem within Ebeltoft Vig.

The three connections to *The Fish* demonstrates exploitation (by some of *The Businesses* and *The Leisure Associations*), dependence (*The Fish's* dependence of *The Sea* along with its water quality and the manmade disturbances happening within the sea), and support (the support given to *The Fish* by some of *The Leisure Associations* by the acts of improving the viability for *The Fish* through underwater habitation places, egg tumblers and fry aquariums at land). The critter that I have dubbed *The Fish's* connections in the *string figure* underlines its role for the cultural environment both in terms of its historical role, its present role and future role. *The Fish* has been of vital economic importance to the fishing harbor, a healthy food source for the people buying it and has also been vital for the community, culture and activities at the harbor. However, in the future management of the heritage site the exploitation of *The Fish* should shift to a more sustainable "interaction" or, as the *string figure* expresses, more symbiotic connection, to ensure an end of the exploitation of *The Fish* and replace it with another type of "interaction". Today, *The Fish* also provides an experience, atmosphere and identity to the heritage site which could be kept alive through experience-based interactions not resulting in a near extinction of parts of *The Fish*.

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3.2. Understanding the village Alling through one significant *string figure* connection

The critters for Alling (Fig. 5 and 6) are noted as: 1) *The Residents*, 2) *The Layout Within The Village*, 3) *The Surrounding Landscape* and 4) *The Buildings*. The village is an old, so-called, "*stjerneudstykning*" which can be translated into "star-shaped allotment plots". In the late 1700 Denmark underwent a large structural reform resulting in changes in the way we cultivated the soil and the ownership of the fields. Before the reform many small villages shared the land and cultivated the soil where it was best suited for the different purposes. To better the individual peasant's right to land most of Denmark was divided into fields with one farm allocated to one piece of land. This resulted in many farming communities being dissolved because the farms were moved out to each piece of land for efficiency purposes. Alling is one of the few villages where the farms remained in a cluster which sustained their community even though their shared land, surrounding the village, was divided into individually owned fields. When viewing the village on a map the farms and small houses are gathered in a cluster surrounded by fields that "shines" (like a star) in lines out from the village. While also visible when visiting the town (Fig. 6), the cadastral division today still reflects the former "starshaped allotment plots" (Fig. 7).

I will focus on the connection between the two critters *The Residents* and *The Surrounding Landscape* which is a connection that has been of vital importance for the local community through the last centuries. Through the years most of the family-owned farms in the village have been transformed into regular family homes and the fields has been rented to large farming companies. The Residents of Alling's current connection to the landscape is thus mainly through recreational walks on designated pathways alongside the fields.



Fig. 5 – *Picture of the village of Alling, Denmark* (Kirkegaard, 2024).



Fig. 6 – *Picture of fields surrounding the village of Alling, Denmark* (Kirkegaard, 2025).



Fig. 7 – *The illustration shows an orthophoto of Alling from 2024. The red lines are the current cadastral division and they are almost true to the division of land made in 1782*. (Plan- og Landdistriktsstyrelsen, 2024).

In 2024 a group of residents formed an association with the aim of transforming one of the fields surrounding the village into a wild nature reserve. They contacted one of the owners which donated his field to the project. The rewilding project strives to benefit biodiversity, but also form a voluntary co-creation project for the residents of Alling. Four yearly meetings have been planned in the village's community house and the residents of Alling are invited to become ambassadors for the future plants and

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animals inside the rewilding zone. For example, one resident could become responsible for a type of flower by counting and monitoring it and another resident could keep an eye out for the emergence of some rare insects (Kirkegaard, 2025). In other words, the rewilding project strives to form a new interaction with the landscape. An interaction that is not merely spectacle by recreational walks, but an active interaction similar to the active interaction that previously existed between the village farms and the fields. However, this new interaction is intended to benefit the natural balance of nature and not merely exploit it.

The *string figure* connection, that I have enhanced, between *The Residents* and *The Surrounding Landscape,* represents a connection that, historically, has been of vital importance for the village’s livelihood and sense of community. It is a great part of the narrative and DNA of the village. By reforming the connection into a new symbiotic connection, the village is reviving a part of its identity and supporting the community feeling through the collective rewilding project. Thus, the connection between *The Residents* and *The Surrounding Landscape* represents an important understanding of the situatedness within the landscape and the collective interaction with the soil and the plants – both today, in the past and its role in the future.

3.3. The unfortunate development of the heritage site *Godsbanen*

Godsbanen, as it is called, is a former freight rail area near the city centre of Aarhus (Aarhus Kommune 2013, pp. 220–223). It was an industrial working zone for the Danish state lines (DSB), but in 2000 DSB vacate the area and the municipality buys the northern part of the area. Due to the financial crisis in 2008 development plans came into a standstill. The area was fenced off and consisted of a wide, open space with gravel, a few bushes and trees, train tracks and a few vacant buildings and containers.

Around 2009 a creative group of young people were allowed by the municipality to occupy some of the buildings with the aim to create a small cultural centre. After a couple of years, the creative group, called Bureau Detours, managed to create a space for alternative exhibitions and events, and Godsbanen became an alternative offer for young people of Aarhus to express themselves. Eventually, small business start-ups and creative initiatives gathered at the site, in the containers, or inside the buildings (Figs. 8 & 9). The activities at the site were gathered under one association called “Institute for X”. To this day Institute for X still exists and foster several start-ups, creative groups and functions as a creative community.



Fig. 8 – *Picture of one of the former railway tracks surrounded by colourful sheds/containers inhabited by the creative association Institute for X, Godsbanen, Aarhus, Denmark* (Kirkegaard, 2022).



Fig. 9 – *Picture of one of the historical, railway buildings inhabited by Institute for X, Godsbanen, Aarhus, Denmark* (Kirkegaard, 2022).

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A contrasting difference between 2009 and now is the surrounding site to Institute for X. In 2009 the creative environment was left undisturbed at the vast industrial site. They could use the materials found at the site and they were allowed to use many of the vacant buildings. The landscape surrounding them was an open space with train tracks stretching into the horizon. As one of the members of Institute for X described it:"It was like a dream space where you could see to the end of the world*"* (Kirkegaard et al., 2021).

The Godsbanen is located in the former river valley which makes the space uniquely flat and open in the otherwise dense and hilly town of Aarhus. Today Institute for X only has a small area and a few buildings to "be creative" within. The site is otherwise (2025) more or less packed with new developments (Fig. 10): Apartment blocks, a parking garage and office buildings. Aarhus School of Architecture moved into a new building at the site in 2021. The municipality transformed one of the historical buildings into a 'Culture House' with concert hall, work spaces and event spaces in 2012 (Fig. 11). When analysing the historical, urban site of Godsbanen through the *string figure* perspective I can divide the site into the following nine critters: 1) *The Former River Valley*, 2) *The Insects And Plants*, 3) *The Municipality's 'Culture House'*, 4) *The New Developments*, 5) *The Historical Buildings*, 6) *Institute for X*, 7) *Residents and Employees/Students*, 8) *Guests* and 9) *Platform 7* (which is a safe space for homeless people). These categories represents both people at the site, the buildings and the landscape.

The former river valley has a surprisingly rich plant and insect life which has a high level of biodiversity, but a lot of it has been destroyed by the new developments. *The New Development* is, in the *string figure* analysis, not connected to *The Former River Valley* or *The Insects And Plants*. Likewise, *The New Development*, is not connected to *The Historical Buildings* or made with respect to the historical urban site as a whole.

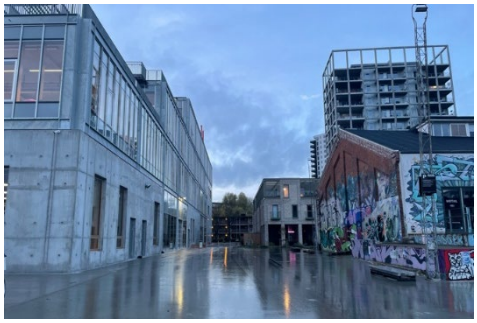


Fig. 10 – *Picture of a part of the heritage site Godsbanen where concrete pavement has replaced the former gravel and train tracks. New buildings are sourrounding the hertage buildings which is seen in the right side of the picture with red bricks and graffiti. Godsbanen, Aarhus, Denmark* (Kirkegaard, 2022).

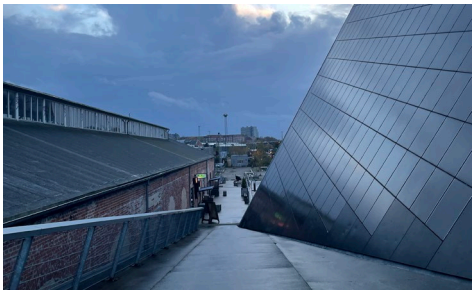


Fig. 11 – *Picture of the Municipality's 'Culture House'. A transformed historical building now in use as a cultural centre with concert hall, work spaces and event spaces. Godsbanen, Aarhus, Denmark* (Kirkegaard, 2022).

Some of the new developments have demolished historical buildings, removed historical train tracks and significant biodiversity. Furthermore, the site's historical relevance could be found in the vastness and openness of the area which today, with the new constructions, has been densified and lost.

The creative group, Institute for X, has, in contrast to *The New Development*, managed to transform the historical buildings and left spots for biodiversity to flourish with the mantra of not cutting any (of the few) trees or bushes down. Together with the municipality owned 'Culture House', which is located in two former industrial buildings, the Institute for X is the only other

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actor at the site that preserves some of the heritage characters. This is placing *Institute for X* with many positive connections within the *string figure.*

Institute for X has, unintentionally, become one of the only actors that preserves the existing buildings, nature, train tracks, etc. The spirit of the initiative Institute for X is an ad hoc, slow development within the existing, given “frames” (buildings, materials, landscape). The new developments demolishes both heritage and landscape and has formed no connections in the *string figure* perspective. In contrast the connections Institute for X has formed to the sites critters exemplifies a sensitive, site-specific development with regards to Aarhus’ collective railway history.

4. Conclusions

The *string figure* perspectives that I have used to analyze and describe the three heritage sites of this paper are a subjective method. The findings would be different if another researcher with a different professional background mapped the critters and the connections. However, the perspectives emphasize the need for a more detailed understanding and management of the heritage sites and the intertwined and essential connections.

Donna Haraway describes how *string figures* are like “stories; they propose and enact patterns for participants to inhabit, somehow, on a vulnerable and wounded earth” (Haraway, 2016, p. 10). At the three sites the *string figure* perspective illustrates how the connections that are formed and the connections that are lacking indicate where the site is experiencing faults. Haraway describes how she is not interested in restoration, but in the more modest possibility of merely “getting on together” (Haraway, 2016, p. 10). At both Godsbanen, Alling and Ebeltoft Fishing Harbor there are connections between critters that could be optimized towards a more balanced relationship. For example, how would the site of Godsbanen have been developed if the new development had considered “getting on together” with the insects and plants and the historical buildings?

Haraway sees an urgency for us to consider the world as an interconnected web spun between microbes, plants, animals, humans, machines, things, nonhumans, etc. and this view is also relevant in relation to heritage sites which often represent deep, historic, many-layered places with complex connections that both form the heritage site’s narrative today and direction for future management. By using the *string figure* perspective, it is possible to shed light on hitherto undetected or under-prioritized connections that could help form a more sustainable future.

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